

Two-Stage High Power Rocket

2026 Argonia Rocket Cup — Completed

Design & analysis: MATLAB, Excel, OpenRocket | Fabrication: fiberglass composite layup, 3D printing | Avionics: redundant multi-computer recovery & staging | Team lead, 6th of 21 teams

Role - Argonia team lead

As team lead, I led the structures sub-team, oversaw the avionics and payload sub-teams, and directed the complete development of both the competition and proof-of-concept rockets.

Overview

As the Argonia team lead for the MSOE High Powered Rocketry Club, I directed the design and manufacture of a two-stage high power rocket for the 2026 Argonia Rocket Cup, the club's most sophisticated vehicle to date. The rocket was designed to reach a target apogee of 16,000 ft while carrying a payload of 18 golf balls, for a projected competition score of 57,000 points. The team was organized into three sub-teams: structures, avionics, and payload. I led the structures sub-team and oversaw the avionics and payload sub-teams, and the rocket placed 6th out of 21 teams.

Proof of Concept – Mini Argonia

Before committing to the full-scale competition vehicle, the team designed and flew a half-scale two-stage rocket during the fall semester, using the same electronics and recovery configuration as the full-scale design.

The design of the proof-of-concept rocket was led by the constraints of the full-scale vehicle, so that it would serve as a faithful, low-cost testbed. It was held to the same 3-inch body tube diameters as the competition rocket and reused the same ebays, allowing the staging and recovery electronics to be validated in their final configuration. It was made dual deployment to mirror the recovery architecture of the full-scale design and was built from easy and inexpensive materials such as cardboard and plywood to keep costs low and turnaround fast. Finally, the vehicle was kept within the Level 1 impulse range to simplify motor certification and launch logistics, pairing an H100 in the sustainer with an H550 in the booster.

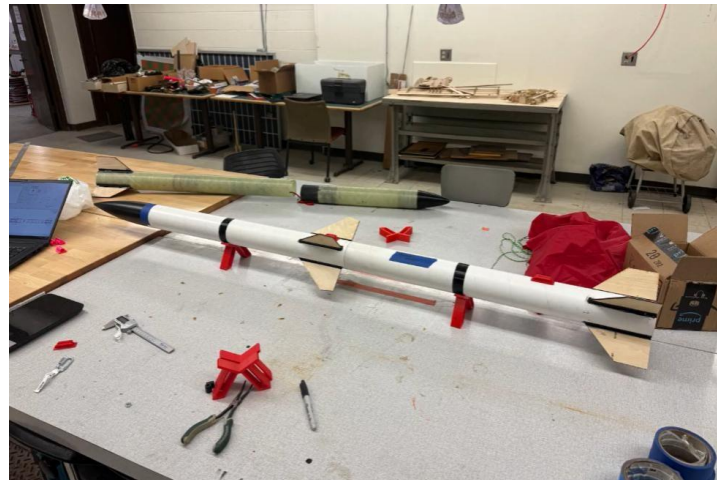


Figure 1 Mini Argonia

At the first launch attempt, the sustainer failed to ignite due to improper programming of the staging logic. Ignition was gated on time, velocity, and altitude: the sustainer would only fire if it detected that it was at least 150 m above the pad, 2.5 seconds after launch, and traveling at least 60 m/s. However, the logic that started the flight timer was miscoded so that it never began counting when launch was detected, so none of the staging conditions were ever evaluated and the sustainer never ignited. All of the dual-deployment recovery logic fired correctly, and the rocket was recovered with massively helpful flight data, making the launch a success for data collection and recovery testing and allowing us to

ensure a fully functional second flight. The second flight succeeded and the sustainer flew to an apogee of 3003 ft AGL and recovered successfully with minor cosmetic damage. This became the first successful two-stage rocket flight in the university's history, establishing a flight-proven staging and recovery configuration for the competition rocket. With the second flight validating the staging logic, electronics, and recovery system end to end, the team was confident moving forward and began the design of the full-scale competition rocket.

Structures

The airframe uses 3-inch fiberglass body tubes, chosen for their high strength-to-weight ratio and their ability to survive the loads of a Mach-1 flight without the added mass of a heavier composite layup. Both stages use 1/4-inch plywood fins; plywood was chosen for its low cost, ease of fabrication, and adequate stiffness given the rocket's comparatively low maximum velocity, avoiding the weight and expense of composite fins that the flight regime did not require. Because the rocket's maximum velocity is comparatively low for its size (Mach 0.9), a full structural tip-to-tip layup was judged unnecessary. Instead, each pair of fins received a single tip-to-tip layup of fiberglass: rather than providing meaningful structural reinforcement, this layer was added to seal the fin-to-body joint, reduce cosmetic damage, and prevent chipping in the event of a hard ground recovery, trading a small amount of added mass for improved durability and reusability.



Figure 2 In progress fiberglass layups

Motor selection was driven directly by the competition scoring equation rather than by apogee alone. The scoring function was modeled in MATLAB and Excel and optimized against its two dominant variables, total impulse and golf-ball count, so that the team could reason about the trade-off between flying higher and carrying more payload mass instead of guessing at it. This analysis defined the impulse budget the vehicle should target; candidate motor pairs within that budget were then modeled in OpenRocket, where predicted apogee, stability margin, and maximum velocity were compared across configurations until the highest-scoring pair that still closed structurally and aerodynamically was identified. The final design pairs a K270 in the sustainer with a K1100 in the booster, for a combined total impulse of roughly 3,556 N-s, a simulated apogee above 16,000 ft, and a maximum velocity of ~Mach 0.9. The combination was selected because it maximized the scoring equation while keeping peak velocity low enough to avoid the mass penalty of a heavier high-Mach airframe.



Figure 3 Aerotech Motor Casings

The rocket is 112 inches long overall, split into a 73.4-inch sustainer and a 42-inch booster, with the length distribution set to place the center of gravity ahead of the center of pressure for adequate stability margin at both the stacked and separated configurations. The two stages are joined by a fiberglass interstage coupler recessed within the booster, which houses a separation charge that fires at booster burnout; recessing the coupler protects the electronics and keeps the separation plane clean so the stages' part reliably a proven design from the proof-of-concept rocket. Each stage uses independent dual deployment so that a failure on one stage cannot compromise recovery of the other: the booster carries an 18-inch drogue and a 36-inch main, and the sustainer a 24-inch drogue and a 50-inch main. Parachute sizes were selected to hold descent under drogue fast enough to limit drift yet keep touchdown velocity within a safe range, and they were deliberately sized so that recovery remains safe even in the off-nominal case where the

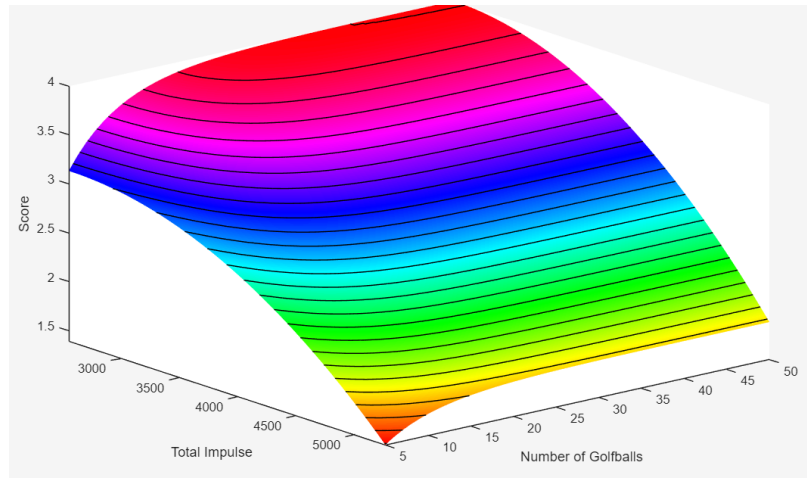


Figure 4 3D plot of Score Equation

sustainer motor fails to ignite and the full stack comes down together. The flight sequence reflects the same fail-safe philosophy: at booster burnout the separation charge fires and the vehicle coasts for 0.5 seconds before sustainer ignition, giving the stages time to separate cleanly and the ignition logic time to confirm conditions; if those conditions are not met, ignition is locked out and the booster simply coasts to apogee and deploys its drogue, ensuring a controlled recovery in every branch of the flight.

Avionics

The avionics design was shaped directly by lessons learned from the previous year's attempt, in which poor electronics choices led to a complete loss of the rocket. In that flight the flight computers were armed using momentary limit switches, which offered quick and easy arming on the pad by simply removing a safety pin but proved incredibly susceptible to vibration. Under the vibration of powered flight, the switches disconnected, the flight computer lost power, and none of the recovery charges deployed, resulting in the loss of the entire vehicle. To eliminate this failure mode, the new ebays were built around rotary switches, which provide a far more positive, vibration-resistant connection while remaining just as quick to arm on the pad and fitting within the same small form factor. The ebay structure was then redesigned around these switches to integrate them cleanly and guarantee that the power connection could not be broken by flight loads.

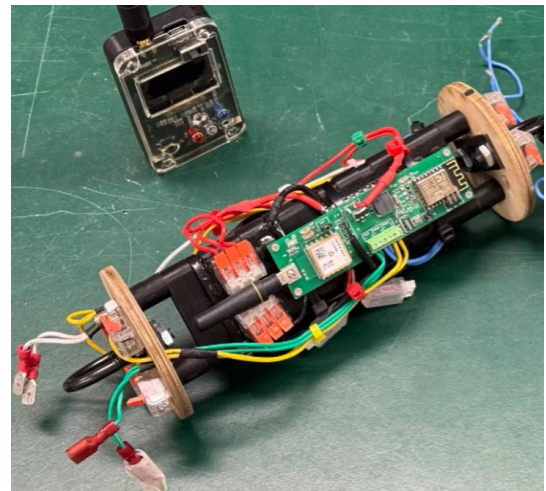


Figure 5 Sustainer Ebay

The entire avionics stack was rebuilt for this vehicle and flight-proven on the half-scale model before being committed to the competition rocket. The booster carries two EasyMini altimeters for fully redundant recovery, chosen for their proven reliability and small form factor. The primary altimeter is programmed directly from the OpenRocket simulation values, while the backup is intentionally staggered to fire just after apogee and just below the target main-deployment altitude. This staggering ensures that a single altimeter failure cannot prevent recovery, while preventing both units from firing their charges simultaneously.

The sustainer carries a Fluctus flight computer as its primary controller, handling drogue deployment, main deployment, and sustainer ignition, backed by an EggTimer Quasar that provides redundant drogue and main deployment and controls booster-to-sustainer separation on its third channel. A dedicated separation event was chosen over hot staging because cleanly separating the stages before ignition removes the risk of the sustainer motor's exhaust damaging the booster and produces a more reliable, repeatable staging sequence. Ignition is time- and altitude-based rather than tilt-based, a decision driven by the previous years' failures with tilt-lockout staging: the sustainer ignites only when velocity exceeds 200 ft/s, the time since liftoff is within two seconds of the planned staging time, and altitude is at least 75 percent of the intended staging altitude. Requiring all three independent conditions to agree confirms the vehicle is flying fast, straight, and high enough before ignition, and if any single condition is unmet the sustainer is locked out, defaulting to a safe recovery rather than an unsafe ignition.

Payload

The rocket carries all 18 golf balls in its forward section, split between the nose cone and the upper sustainer body directly beneath it. Locating the payload this far forward was a deliberate design decision rather than a packaging convenience: concentrating mass ahead of the center of pressure pulls the center of gravity forward, which directly improves the stability margin and let the team trim fin size and drag rather than adding ballast or oversized fins to stay stable. Consolidating the payload into the nose and upper sustainer also shortened the overall airframe, reducing structural mass and drag and freeing a clean, well-supported bulkhead at the base of the payload bay for a stronger recovery-harness attachment.

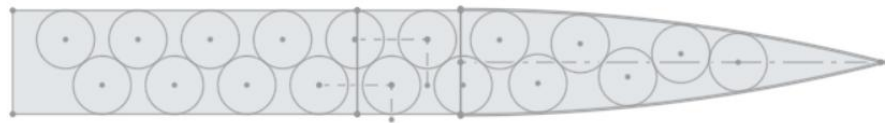


Figure 6 Sketch of payload arrangement

The internal arrangement was optimized for packing efficiency and reliable loading. Golf balls in the sustainer body are stacked in a vertical zigzag and retained laterally by a lightweight 3D-printed track, a layout the team found packed the most balls into the shortest length while keeping them captured against flight loads. Those in the nose cone sit in a lightweight 3D-printed shell shaped to the internal

nose geometry, which keeps them aligned along the center line, preserves the forward mass distribution, and simplifies loading on the

pad. Compared with the previous year's payload, this configuration is simpler, lighter, and denser: the same golf-ball count is carried in less length and mass, and the findings from iterating on the packing scheme fed directly back into the airframe length and stability budget used to size the rest of the vehicle.

Launch and Reflection

The competition launch was a success. At the 2026 Argonia Rocket Cup the rocket lifted off, staged cleanly, ignited its sustainer, and recovered under a full dual-deployment sequence exactly as designed, a fully nominal two-stage flight that capped a full year of iterative development. Flown against a projected competition score of roughly 57,000 points, the vehicle earned the team a 6th-place finish out of 21 competing teams, the club's strongest result to date and its first competition flight built on a flight-proven two-stage staging and recovery configuration.

Reflecting on the project, the result validated the methodical, test-driven approach the team took from the outset. The decision to design and fly a half-scale proof-of-concept vehicle first proved decisive: it surfaced the staging-logic error in a low-stakes flight, let the team validate the electronics and recovery architecture end to end, and built confidence in the configuration before committing to the full-scale rocket. Grounding motor selection in the scoring equation, engineering the avionics around the previous year's failure modes, and sizing every recovery decision for the off-nominal case all contributed to a vehicle that behaved predictably on competition day. Beyond the hardware, the project showed what a student team can accomplish with clear sub-team ownership, disciplined iteration, and a willingness to learn from failure — producing the most sophisticated rocket in the club's history and the first successful two-stage flight in the university's program.

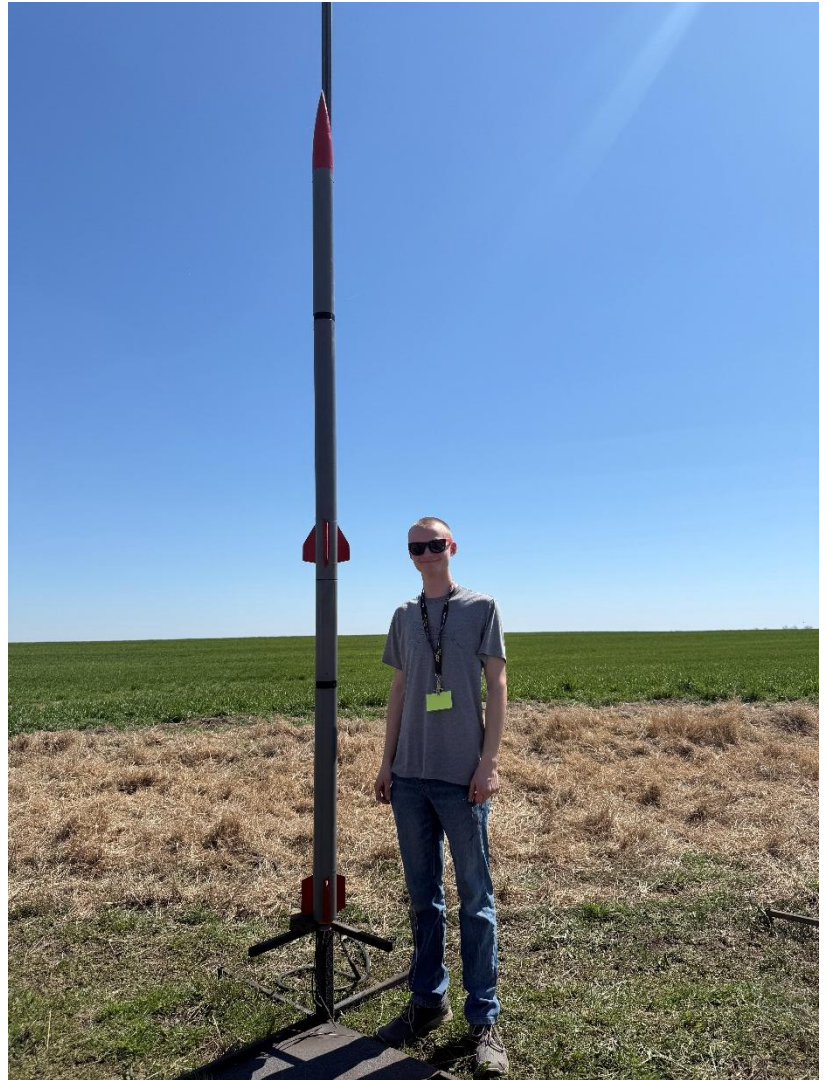


Figure 7 Me next to the rocket on the pad